

Get started with the DESY-SINTEF Beamer Theme

Using $\text{\LaTeX}$ to prepare slides

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Beamer for UIC presentations

A quick start

If you would like $\text{\LaTeX}$ in your presentation, Beamer is a great way to go!

- Beamer has a detailed [user manual](#), but we will go over the most common features.
- This template is designed for a 16:9 aspect ratio, which is the default in PowerPoint and the most common amongst projectors.
- The most common of all slide types involve bulleted points, like these.



Beamer for UIC presentations

A quick start

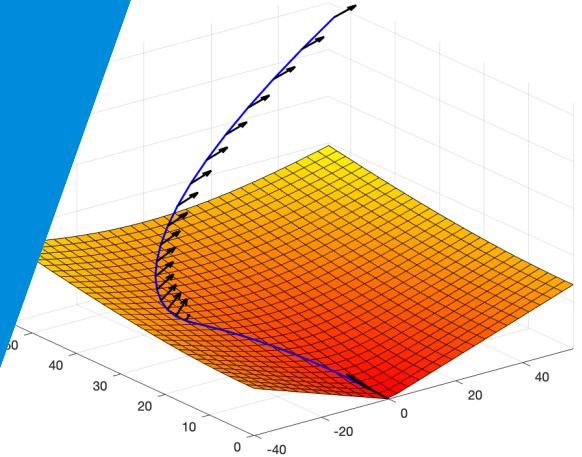
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- The most common of all slide types involve bulleted points, like these.
 - Placing `\pause` after items will allow you to sequentially unroll points.
 - Changing `\begin{itemize}` to `\begin{itemize}[<+>]` achieves a similar effect, although `\pause` gives finer control on the unrolling.



Slide Layouts

Some standard slide layouts





Images, columns and tips for good slides

Do use this optional subtitle where appropriate

- **Do keep** a good balance between text and figures.
 - **Do use** nested lists wherever appropriate.
- **Do use** the `\underbrace` and `\overbrace` commands to help explain your equations:

$$\begin{array}{ll} \max_{\lambda, \nu} . & \underbrace{\inf_x \overbrace{L(x, \lambda, \nu)}^{\text{Lagrangian}}} \\ & \text{Lagrange Dual Function, } g(\lambda, \nu) \\ \text{s.t.} & \lambda \geq 0 \end{array}$$

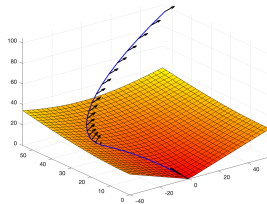


Figure: Minimum fuel trajectory



A **standout** frame with title and subtitle

Subtitle here

A **standout** frame can be used to focus attention



More with bullet points

Alerts and repeats

Sections 12.1 through 12.3 of the [Beamer user manual](#) demonstrate many more features, like alerts and repeats.

- The `\alert{}` feature can be particularly useful.
- Like this.
- The highlighting may span multiple slides in a frame.
 - Previous one did not, but this one will.
- This is accomplished by the hyphen (i.e. -) in the `\alert<3->` command.



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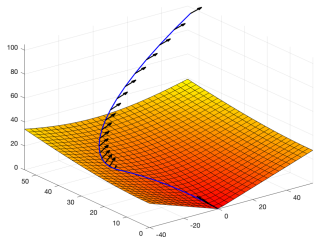
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Another images/columns example



Left: Solving for optimal
fuel consumption **Right:**
Solving for optimal time
taken

Figure: Minimum fuel trajectory

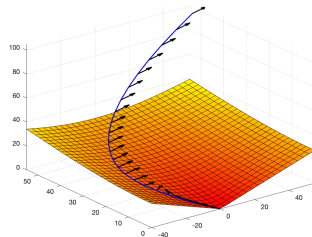


Figure: Minimum time trajectory



Blocks

Some content will just look better in blocks.

Block title

This is a block. Its color will match the color of the footline. The block environment is native to Beamer.

Block title

This Beamer theme also provides a `colorblock` environment which is a wrapper on top of the usual block environment in Beamer. It allows you to specify a custom background and font color.

Here's an example usage:

The discrete case

...

The continuous case

...



Blocks for theorems

Beamer also uses blocks by default to wrap theorems.

Theorem

This is a theorem.



Side-Picture Slides

Subtitle here

This Beamer theme also provides a `sidepic` environment which is a wrapper on top of the `frame` environment.

- It has an optional `image` argument which will help you achieve a layout of this type.





Pseudocode Example

Algorithm Bellman-Kalaba (adapted from [algorithmicx documentation](#)).

Input: G, u, l, p
for all $v \in V(G)$ **do**
 $l(v) \leftarrow \infty$
end for
 $l(u) \leftarrow 0$
repeat
 for $i \leftarrow 1, n$ **do**
 $min \leftarrow l(v_i)$
 for $j \leftarrow 1, n$ **do**
 if $min > \text{Edge}(v_i, v_j) + l(v_j)$ **then**
 $min \leftarrow \text{Edge}(v_i, v_j) + l(v_j)$
 $p(i) \leftarrow v_j$
 end if
 end for
 $l'(i) \leftarrow min$
 end for
 $changed \leftarrow l \neq l'$
 $l \leftarrow l'$
until $\neg changed$

▷ Example comment



Typesetting Algorithms

Common Issues

- There are [multiple environments](#) for typesetting algorithms.
 - This template uses `algorithmicx` via `algpseudocode`.
 - `algpseudocode` is (one of the) layouts for `algorithmicx`.
 - Others exist, as mentioned in the [documentation, section 2.1](#).
 - `algorithmic` is also widely used, which is an alternative to using `algorithmicx` (via a layout package).
- While `algpseudocode` tries to be close to `algorithmic`, they use different capitalization schemes
 - `algorithmic` commands are fully capitalized (e.g. `\STATE`, `\FORALL`)
 - `algpseudocode` commands are not (e.g. `\State`, `\ForAll`).
 - **If moving an algorithm from one to the other, you will need to fix the capitalization, or the compiler will run into problems.**



Typesetting Algorithms

More control

- If you must use `algorithmic` instead of `algpseudocode`, some commands will need to be manually defined
 - For example the function `\Call` (or rather, the `\CALL`) command.
 - Do it as `\newcommand*\CALL[2]{\textsc{#1}(#2)}`
 - Remember that not all fonts have a small caps variant needed by `\textsc`
 - You can look at `uicfont.sty` to see how we augment our font with another small caps font (i.e. **PlayFair Display SC**)
 - Additionally, the `\Comment` of `algpseudocode` might look better than the `\COMMENT` of `algorithmic`
 - You can define a custom one to mimic `\Comment`
 - Do it as `\newcommand*\ANNOTATE[1]{\hfill\(\triangleright\)\ #1}`



Good luck with your presentation!

- This template is hosted on GitHub
 - <https://github.com/SaverioMonaco/DESY-SINTEF>
- I would appreciate contributions of all sorts (pull requests, identifying issues, etc).
- If you have any suggestions, [send them to me!](#)



Thank you!